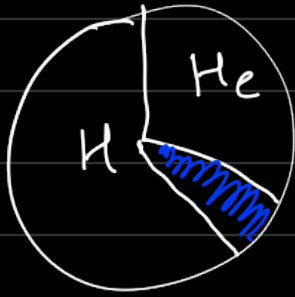


* Metallicity:

- $Z \equiv$ Mass fraction of elements heavier than H & He



- Measure relative to $\odot$

$$Z_{\odot} = \frac{\text{Metals}}{(\text{H} + \text{He})} = 0.02 = \left(\frac{\text{Fe}}{\text{H}} \right)$$

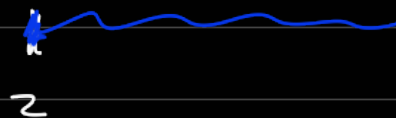
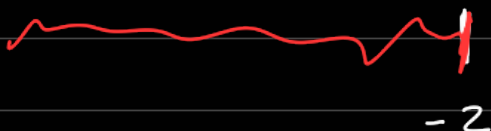
$$\left[\frac{\text{Fe}}{\text{H}} \right] = \log_{10} \left(\frac{Z}{Z_{\odot}} \right)$$

$$\left[\frac{\text{Fe}}{\text{H}} \right] = \log_{10} \left(\frac{\text{Fe}}{\text{H}} \right) - \log_{10} \left(\left(\frac{\text{Fe}}{\text{H}} \right)_{\odot} \right)$$

- Milky way:

$$\left[\frac{\text{Fe}}{\text{H}} \right] \gtrsim +2$$

$$\left[\frac{\text{Fe}}{\text{H}} \right] \lesssim -2$$



* Color-magnitude diagram:

- trigonometric parallax

- No stars nearer than $M_v = -4$ (near (Vamp))

Stellar Luminosity & Mass f^{ns}:

• Present day stellar luminosity function

$$\Phi(M_V) \quad \text{at mag } M_V$$

$$\Phi(M_V) = \frac{N_{\star}}{M_V V}$$

$$\Phi(M_V) \Delta M$$

• $\int \Phi(M_V) \Delta M$: $\left(\frac{\text{total lum. } L}{\text{Vol}} \right)$

Initial stellar Lum. fⁿ:

$$\Psi(M_V)$$

$$\Psi(M_V) = \begin{cases} \Phi_{MS}(M_V) & , \tau_{MS}(M_V) \geq \tau_{gal} \\ \Phi_{MS}(M_V) \cdot \left(\frac{\tau_{gal}}{\tau_{MS}(M_V)} \right) & , \tau_{MS}(M_V) \leq \tau_{gal} \end{cases}$$

$\tau_{MS}(M_V)$: Main seq. lifetime

τ_{gal} : Age of Milky-Way

Initial Mass fⁿ: $\xi(M) \Delta M$

the number of stars born with masses between M and $M + \Delta M$ within some volume V .

$$\xi = \frac{N}{V M}$$

$$\propto \frac{1}{M} \propto M^{-1}$$

$$\xi(M) = \xi_0 (M_0)$$

• Near Sun: $\alpha = 2.35$

$$\xi \Delta M : \frac{\#}{\text{Vol}}$$

Problem sheet:

① $\frac{L_1 D_0^2}{L_0 D_1^2} = (100)^{\frac{1}{5}} (m_0 - m_1)$

$$\frac{L_1 D_0^2}{L_0 D_1^2} = 10^{\frac{2}{5}} (m_0 - m_1)$$

$$L_0 = L_{\odot}$$

$$\frac{L_V}{L_0} = 10^{\frac{2}{5}} (M_{V0} - M_V)$$

$$\frac{L_V}{L_0} = 200 \cdot 10^{\frac{2}{5}} (4.83 - 3.5) + 20 \cdot 10^{\frac{2}{5}} (4.83 - 0.7)$$

②

$$\left[\frac{Fe}{H} \right] = \log_{10} \left(\frac{Z}{Z_0} \right) = -0.03$$

$$Z = 0.0178$$

$$Z_0 = 0.02:$$

③ Near galactic center:

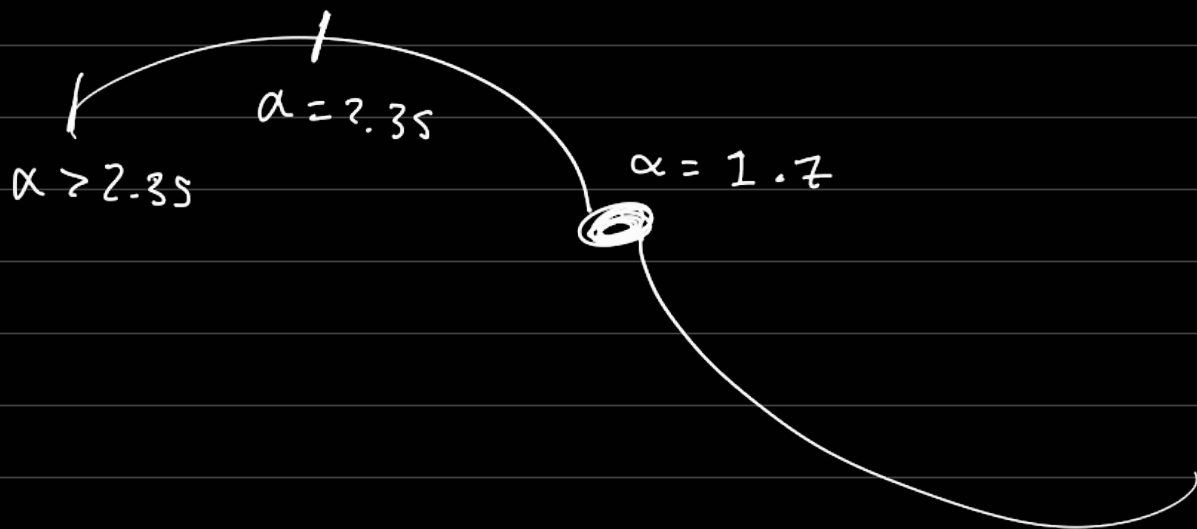
$$\alpha = 1.7$$

$$\xi(M) = \xi_0 M^{-1.7}$$

total: $N_{\text{tot}} = \int_{M_0}^{M_f} \xi(M) dM$

$$\frac{N(M \geq 3M_0)}{N_{\text{tot}}} = \frac{\int_3^{150} \xi(M) dM}{\int_{0.5}^{150} \xi(M) dM} =$$

⊙ Salpeter $\alpha = 2.35$ (near sun)



* Mass contained in dM is $M \xi(M) dM$

$$\langle f(M) \rangle = \int_{0.5}^{150} f(M) M \xi(M) dM$$

$$= \int_{0.5}^{150} M \xi_0 M^{-2.5} dM$$

$$= \xi_0 \int_{0.5}^{150} M^{-1/2} dM$$

$$= \xi_0 2 M^{1/2} \Big|_{0.5}^{150}$$

$$= 2 \xi_0 (\sqrt{150 M_0} - \sqrt{\frac{1}{2} M_0})$$

$$= 2 \xi_0 \sqrt{M_0} (\sqrt{150} - \sqrt{\frac{1}{2}})$$

$$= 2 \xi_0 \int_0^1 \sqrt{M_0} \left(5 \sqrt{2} \sqrt{3} - \sqrt{2}/2 \right)$$

$$= 2 \xi_0 \int_0^1 \sqrt{2M_0} \left(5 \sqrt{3} - 1/2 \right)$$

$$\int_{0.5}^{1.5} M \xi_0 M^{-2.5} dM = 1$$

Solve for ξ_0 and plug in.

Stellar Clusters:

• Form from the same gas:

• Same age

• Same metallicity

↳ open

↳ glob

ISM: phases in MW

• ISM $\approx$ 10% total mass of MW

• Neutral Gas:

• Spiral arms

• Molecular Gas: H_2

• UV light breaks



• CO: second most abundant

$$* n(R, z, s) = n(0, 0, s) e^{-R/h_R(s)} e^{-|z|/h_z(s)}$$



Stellar Motion:

- trigonometric parallaxes:
- Distance using spectral types (photometric parallax)

- Radial Velocity:

Doppler shift of absorption lines

- tangential velocity:

[Proper motion μ
Distance d]



$$V_t (\text{km/s}) = 4.74 d(\text{kpc}) \mu (\text{milliarcsec/yr})$$

•

$$V = \sqrt{V_r^2 + V_t^2} \quad (\text{total vel.})$$

- Velocity dispersion:

$$\sigma_v^2 = \langle V^2 - \langle V \rangle^2 \rangle$$

start with

$$\Delta V = V - \langle V \rangle$$

$$\langle \Delta V \rangle = 0$$

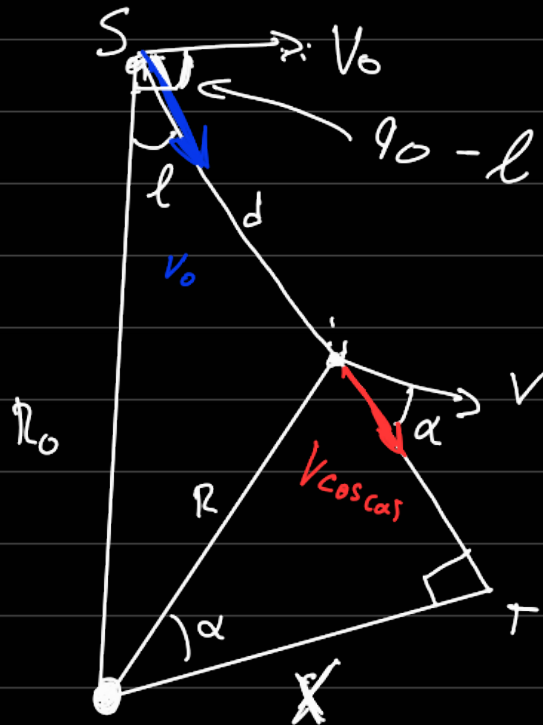
$$\begin{aligned} \Delta V^2 &= V^2 - 2\langle V \rangle \langle V \rangle + \langle V \rangle^2 \\ &= V^2 - \langle V \rangle^2 \end{aligned}$$

$$\langle (\Delta V)^2 \rangle \equiv \sigma^2 = \langle V^2 - \langle V \rangle^2 \rangle$$

* local standard of Rest: (LSR) $\langle V \rangle$

$$\sqrt{\sigma^2} = V_{rms}^*$$

Galactic Rotation:



$$V_r = V \cos(\alpha) + V_0 \cos\left(\frac{\pi}{2} - l\right)$$

$$V_r = V \cos(\alpha) - V_0 \sin(l)$$

$$\cos(\alpha) = \frac{x}{R}, \quad \sin(l) = \frac{x}{R_0}$$

$$\cos(\alpha) R = R_0 \sin(l)$$

$$\cos(\alpha) = \frac{R_0}{R} \sin(l)$$

$$V_r = \sin(l) \left(\frac{R_0 V}{R} - \frac{V_0 R}{R_0} \right)$$

$$V_r = R_0 \sin(l) \left(\frac{V}{R} - \frac{V_0}{R_0} \right)$$

angular velocity $\omega = \frac{V}{R}$

$$V_r = R_0 \sin(\ell) (\omega - \omega_0)$$

$$V_t = (\omega - \omega_0) R_0 \cos(\ell) - V \frac{d}{dR}$$

$$\bullet \omega(R - R_0) = \omega(R_0) + \frac{d\omega}{dR} (R - R_0) + \mathcal{O}(R - R_0^2)$$

$$\omega - \omega_0 = \frac{d\omega}{dR} (R - R_0)$$

$$V_r = \frac{d\omega}{dR} (R - R_0) R_0 \sin(\ell)$$

$$R - R_0 \approx d \cos(\ell)$$

$$V_r = A d \sin(2\ell)$$

$$V_r$$

$$A = -\frac{R_0}{2} \left. \frac{d\omega}{dR} \right|_{R_0} = \frac{1}{2} \left(\frac{V_0}{R_0} - \left. \frac{dV}{dR} \right|_{R_0} \right)$$

$$V_t = d(4 \cos(2\ell) + B)$$

$$B = -\frac{1}{2} \left(\frac{V_0}{R_0} + \left. \frac{dV}{dR} \right|_{R_0} \right)$$

$$V_t$$

$$\omega_0 = \frac{V_0}{R_0} = A - B$$

$$\frac{dV}{dR} = -(A + B)$$

- Use 2l can be Hydrogen
- $R < R_0$

$$V_{r, \max}(\ell) = (\omega(R) - \omega_0) R_0 \sin(\ell)$$

$$R - R_0 \approx d \cos(\ell)$$

$$r = r_0 \sin \alpha$$

tangent point method:

$$V(R) = V_{r, \text{max}}(R) + V_0 \sin(\alpha)$$

